
SUMMARY

- Research scientist (Ph.D. Applied Math) who turns real-world problems into mathematical models, then code, then measurable results across domains including astrophysical flows, networked systems, and industrial operations; published in numerical analysis, networks, and scientific ML.
- Translate physics and operations into solvable equations and find efficient algorithms.
- Collaborative across disciplines (physics, engineering, operations); communicate results clearly.
- Focus: PINNs for PDEs, time-series/simulation modeling, spectral/compressive methods, solid mechanics, and practical signal processing. Python/JAX/MATLAB; Tests and documentation.

SKILLS

Core	Mathematical modeling, numerical methods, stochastic processes
ML & Stats	Time-series, classification/regression, signal processing
Programming	Python (NumPy, pandas, JAX/PyTorch, scikit-learn), MATLAB
Tools	Git, Linux, Slurm, TensorBoard, CVX/MOSEK, pytest
Languages	Chinese (native), English (fluent)

EDUCATION

Ph.D. in Mathematics, University at Buffalo 2016.08 – 2021.08

Thesis: Numerical Solution to the Corner Angle Problem for an Elastic Solid with Anisotropic Surface Energy

B.S. in Mathematics, Fudan University 2011.09 – 2016.06

EXPERIENCE

Postdoctoral Fellow, University of Victoria 2023.11 – present

- **Neural PDE solver:** built a time-stepped neural network model in JAX for accretion-disk equations; reproduced spiral density waves and gap formation; handled periodic and radial boundaries; compared against numerical baselines.
- **Age of Information theory:** derived results for time-varying service systems; computed transient distributions and steady-state summaries.

Postdoctoral Fellow, Concordia University 2021.09 – 2023.05

- **High-dimensional diffusion:** developed a compressed-sensing spectral method for diffusion PDEs; proved correctness guarantees; demonstrated scalability to 20D.
- **Routing for fresh information:** defined an edge-AoI objective; designed two 2-approximation algorithms; validated on large synthetic grids.
- **Railway delay:** built an interpretable Markov forecaster with kernel smoothing on 1M rows; reported clear decision metrics.

Research Assistant, University at Buffalo 2018.01 – 2021.08

- **Solid mechanics:** developed numerical solvers for an anisotropic surface-energy corner-angle problem in elasticity; validated against analytical limits; implemented in MATLAB with parallelization.

Research Intern, Fudan Microelectronics Group 2014.08 – 2016.06

- **Signal processing for side-channel data:** PCA denoising and alignment; template-based classification improved key-recovery accuracy; recognized on DPA v4 Hall of Fame.

SELECTED PUBLICATIONS

- Approximation schemes for Age of Information minimization in UAV grid patrols (2025).
- Neural Networks as Surrogate Solvers for Time-Dependent Accretion Disks (submitted 2025).
- Compressive Fourier collocation methods for high-dimensional diffusion (2024).
- Does elastic stress modify the equilibrium corner angle? (2022).